

Desflurane is not inferior to sevoflurane in the occurrence of adverse respiratory events during laryngeal mask airway anesthesia: a non-inferiority randomized double-blinded controlled study.

PMID: 32013332 | Indexed in PubMed

Compared to sevoflurane, desflurane is less favorable to most anesthesiologists for laryngeal mask airway (LMA) anesthesia because desflurane has a pungent odor. This non-inferiority study aimed to determine whether desflurane is not worse than sevoflurane in triggering airway irritations during general anesthesia using LMA. Patients scheduled for elective surgery under LMA anesthesia were randomly allocated to receive either desflurane or sevoflurane for maintenance of anesthesia. After intravenous fentanyl, lidocaine and propofol administration followed by LMA insertion, 0.5-1.0 MAC of the volatile anesthetic in 50% N₂O was maintained throughout the surgery. The primary outcome was the occurrence of perioperative adverse respiratory events. Other outcomes included recovery profiles, hemodynamic changes and postoperative complications. One-hundred and ten patients per group completed the study without any serious complications, lost to follow-up, or protocol deviation. During awake LMA removal, patients in the desflurane group experienced lesser episodes of laryngospasm (risk difference, -7.3%; 95% CI, -12.7% to -1.9%; P=0.009) than those in the sevoflurane group. The emergence time and time to LMA removal were significantly shorter in the desflurane group. The quality of recovery indicated by an ability to self-transfer from bed to bed was significantly better after desflurane anesthesia. No difference between groups was found in a return of orientation and a readiness for post-anesthesia care unit discharge. Desflurane is non-inferior to sevoflurane in the occurrence of laryngospasm at emergence after LMA anesthesia. The superiority of desflurane compared to sevoflurane with regards to respiratory complications requires further investigation.

Comparison of the Occurrence of Postoperative Shivering Between Sevoflurane and Desflurane Anesthesia.

PMID: 35049406 | Indexed in PubMed

General anesthetic agents can change the shivering threshold. Sevoflurane and desflurane are widely used as inhalational anesthetics and have also been reported to lower the shivering threshold in a dose-dependent manner. Although the comparison of postoperative shivering (POS) between total intravenous anesthesia and inhalational anesthesia has been reported, there have been no reports on a direct comparison between sevoflurane and desflurane anesthesia and the occurrence of POS in open abdominal surgery. After obtaining approval from the Ethics Review Committee (2020-261), 683 adult patients who underwent open radical surgery for uterine, cervical, or pancreatic cancer under general anesthesia using inhalational anesthetics at Kyushu University hospital between December 2012 and March 2020 were included in this retrospective study. The odds ratio (OR) for the occurrence of POS between the two groups (sevoflurane and desflurane) was calculated. Multivariable-adjusted analysis was performed using possible factors affecting POS. Furthermore, propensity score (PS) matching was conducted using these factors. The multivariable-adjusted OR for the occurrence of shivering in the desflurane group (62 occurrences/356 patients) was 1.06 (95% confidence interval [CI]: 0.69-1.62, p = 0.79) compared with the sevoflurane group (77/327, reference). Similarly, after PS matching, the crude OR for the occurrence of shivering in the desflurane group (47/210) was 1.09 (95% CI: 0.68-1.75, p = 0.72) compared with the sevoflurane group (44/210, reference). Similar results were obtained in the stratified analysis by sex and age. The occurrence of POS is not different between sevoflurane and

desflurane anesthesia.

Recovery of postoperative cognitive function in elderly patients after a long duration of desflurane anesthesia: a pilot study.

PMID: 25638572 | Indexed in PubMed

Postoperative cognitive dysfunction (POCD) increases morbidity and mortality. The mechanisms underlying POCD remain elusive; however, systemic responses induced by anesthesia and surgery might trigger neuroinflammation and POCD. Desflurane is a preferable volatile anesthetic agent for elderly patients because it facilitates shorter recovery from general anesthesia. The aim of this study was to determine whether quality of emergence and cognitive function in elderly patients undergoing a long duration desflurane anesthesia are better than those in the case of sevoflurane anesthesia. Forty-two patients who were older than 65 years of age and scheduled for surgery of more than 4 h in duration were enrolled in this study. Patients were randomly assigned to a desflurane anesthesia group (D group) and sevoflurane anesthesia group (S group). General anesthesia was maintained with 3.5 % desflurane (D group) and 1.0 % sevoflurane (S group). The Mini-Mental State Examination (MMSE) was used for assessing cognitive function 24 h before and after surgery. Postoperative MMSE score in the D group was significantly improved compared to that in the preoperative period. In conclusion, elderly patients undergoing desflurane anesthesia have significantly better quality of emergence and may have better cognitive function than those in elderly patients undergoing sevoflurane anesthesia.

Experiences with the new inhalational agents in low-flow anesthesia and closed-circuit technique. Monitoring and technical equipment.

PMID: 10150774 | Indexed in PubMed

During recent years interest has focused on two completely fluorinated ethers, desflurane and sevoflurane, which promise a shorter induction of and emergence from anesthesia. Their physicochemical properties differ from isoflurane, enflurane and halothane, thus requiring new technical equipment and leading to a change in anesthesiological procedures. Low-flow anesthesia with desflurane can be performed, the technical equipment is available, especially vaporizers and gas analyzers. In contrast to anesthesia with isoflurane, enflurane and halothane, the initial high-flow wash-in period with desflurane can be shorter and the vaporizer setting can remain unchanged after fresh gas flow reduction. In order to administer desflurane and sevoflurane in closed circuit technique, new technical equipment is needed. Therefore, a computer controlled anesthesia machine was modified and the feedback mechanism to maintain the end-tidal anesthetic concentration was simulated. Isoflurane, desflurane or sevoflurane needed the same time for wash-in. Wash-out was slower with isoflurane; however, the technical equipment should be adapted to increase the elimination of the new agents. The consumption of desflurane and sevoflurane is effectively reduced by low-flow and closed circuit anesthesia.

[Desflurane: a new inhalation anesthetic].

PMID: 8966353 | Indexed in PubMed

Desflurane is a new fluoride-only halogenated inhalational anesthetic. It differs from other halogenated anesthetics, in having a lower solubility in all tissues and greater molecular stability in all media. These traits mean, on the one hand, that desflurane affords rapid achievement of depth of anesthesia, recovery and management during surgery, and on the other, that its potential toxicity is low and that it is safe for use in low flow circuits. The pharmacodynamic effects of desflurane are similar to those of isoflurane in the blood stream, airways and brain. Sympathetic hyperactivity with increased heart rate and arterial pressure may be triggered when concentration is increased quickly, but this effect is partially abolished when fentanyl is administered. Induction of anesthesia with desflurane causes irritation of the airways, particularly in children, and its use in pediatric surgery is therefore inadvisable. Desflurane potentiates the action of neuromuscular relaxants, much like isoflurane. Desflurane represents a step forward in the search for the ideal anesthetic.

The Role of GABA Receptors in Anesthesia and Sedation: An Updated Review.

PMID: 39465449 | Indexed in PubMed

GABA (γ -aminobutyric acid) receptors are constituents of many inhibitory synapses within the central nervous system. They are formed by 5 subunits out of 19 various subunits: $\alpha 1-6$, $\beta 1-3$, $\gamma 1-3$, δ , ϵ , θ , π , and $\rho 1-3$. Two main subtypes of GABA receptors have been identified, namely GABAA and GABAB. The GABAA receptor (GABAAR) is formed by a variety of combinations of five subunits, although both α and β subunits must be included to produce a GABA-gated ion channel. Other subunits are γ , δ , ϵ , π , and θ . GABAAR has many isoforms, that dictate, among other properties, their differing affinities and conductance. Drugs acting on GABAAR form the cornerstone of anesthesia and sedation practice. Some such GABAAR agonists used in anesthesia practice are propofol, etomidate, methohexital, thiopental, isoflurane, sevoflurane, and desflurane. Ketamine, nitrous oxide, and xenon are not GABAAR agonists and instead inhibit glutamate receptors-mainly NMDA receptors. In spite of its many drawbacks such as pain in injection, quick and uncontrolled conversion from sedation to general anesthesia and dose-related cardiovascular depression, propofol remains the most popular GABAAR agonist employed by anesthesia providers. In addition, being formulated in a lipid emulsion, contamination and bacterial growth is possible. Literature is rife with newer propofol formulations, aiming to address many of these drawbacks, and with some degree of success. A nonemulsion propofol formulation has been developed with cyclodextrins, which form inclusion complexes with drugs having lipophilic properties while maintaining aqueous solubility. Inhalational anesthetics are also GABA agonists. The binding sites are primarily located within $\alpha +/\beta -$ and $\beta +/\alpha -$ subunit interfaces, with residues in the $\alpha +/\gamma -$ interface. Isoflurane and sevoflurane might have slightly different binding sites providing unexpected degree of selectivity. Methoxyflurane has made a comeback in Europe for rapid provision of analgesia in the emergency departments. Pentrox (Galen, UK) is the special device designed for its administration. With better understanding of pharmacology of GABAAR agonists, newer sedative agents have been developed, which utilize "soft pharmacology," a term pertaining to agents that are rapidly metabolized into inactive metabolites after producing desired therapeutic effect(s). These newer "soft" GABAAR agonists have many properties of ideal sedative agents, as they can offer well-controlled, titratable activity and ultrashort action. Remimazolam, a modified midazolam and methoxycarbonyl-etomidate (MOC-etomidate), an ultrashort-acting etomidate analog are two such examples. Cyclopropyl methoxycarbonyl metomidate is another second-generation soft etomidate analog that has a greater potency and longer half-life than MOC-etomidate. Additionally, it might not cause adrenal axis suppression. Carboetomidate is another soft analog of etomidate with low affinity for 11β -hydroxylase

and is, therefore, unlikely to have clinically significant adrenocortical suppressant effects. Alphaxalone, a GABAAR agonist, is recently formulated in combination with 7-sulfobutylether- β -cyclodextrin (SBECD), which has a low hypersensitivity profile.

Comparison of remimazolam and sevoflurane for general anesthesia during transcatheter aortic valve implantation: a randomized trial.

PMID: 39715979 | Indexed in PubMed

Safe perioperative management of patients undergoing transcatheter aortic valve implantation (TAVI) is crucial. Remimazolam is a newly developed short-acting benzodiazepine. We hypothesized that combining remimazolam and flumazenil would reduce emergence time compared with sevoflurane in patients undergoing general anesthesia for TAVI. We conducted a prospective, randomized, parallel-design, open-label, single-centre clinical trial between June 2022 and August 2023 at Kagoshima University Hospital. We allocated patients randomly to either the remimazolam/flumazenil group or the sevoflurane group. Patients in the remimazolam group received iv remimazolam whereas patients in the sevoflurane group received sevoflurane for general anesthesia maintenance. Patients in both groups received a remifentanyl infusion throughout the TAVI procedure ($0.2 \mu\text{g}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ iv). Remimazolam and sevoflurane were adjusted to maintain a Bispectral Index™ (Covidien/Medtronic, Minneapolis, MN, USA) of 40-60. In the remimazolam group, flumazenil (0.2 mg iv) was administered immediately after remimazolam discontinuation. The primary outcome was time to extubation. Secondary outcomes included intraoperative variables (hemodynamic variables and vasopressor dose), rate of intra- and postoperative complications, and recovery of muscle strength. Overall, 60 patients were enrolled, and data from 56 were included. The median [interquartile range] time to extubation was significantly shorter in the remimazolam group than in the sevoflurane group (6.5 [5.1-8.1] min vs 14.2 [10.9-15.9] min; difference in medians, -6.9 min; 95% confidence interval, -8.7 to -5.0; $P < 0.001$). Statistically significant differences were observed in the perfusion index ($P = 0.03$) and regional cerebral oxygen saturation ($P = 0.03$) between the groups. No significant differences between the two groups were seen in other secondary outcomes. Compared with sevoflurane, a combination of remimazolam and flumazenil significantly reduced the time to extubation in patients undergoing general anesthesia for TAVI. Therefore, remimazolam may be a suitable choice for general anesthesia in patients undergoing TAVI. UMIN.ac.jp (UMIN000047892); first posted 30 May 2022.

Anesthetic properties of some fluorinated oxolanes and oxetanes.

PMID: 17456657 | Indexed in PubMed

The search for new potent inhaled anesthetics has slowed, in large part because of the excellence of the two most recent additions, desflurane and sevoflurane. Nonetheless, neither desflurane nor sevoflurane are ideal anesthetics, desflurane causing cardiorespiratory stimulation, and sevoflurane having a slower (albeit rapid) recovery from anesthesia. Sevoflurane also can produce convulsions and postoperative agitation. In the present report, we describe the physical and anesthetic properties of 31 cyclic ethers halogenated solely with fluorine. Although several produced anesthesia, none had solubilities that would make them better than sevoflurane. The remaining ethers were unstable or produced obvious central nervous system irritation, including convulsions. We find that none of these cyclic ethers appear to

provide advantages over desflurane or sevoflurane.

Volatile Anesthetics versus Total Intravenous Anesthesia for Cardiac Surgery.

PMID: 30888743 | Indexed in PubMed

Volatile (inhaled) anesthetic agents have cardioprotective effects, which might improve clinical outcomes in patients undergoing coronary-artery bypass grafting (CABG). We conducted a pragmatic, multicenter, single-blind, controlled trial at 36 centers in 13 countries. Patients scheduled to undergo elective CABG were randomly assigned to an intraoperative anesthetic regimen that included a volatile anesthetic (desflurane, isoflurane, or sevoflurane) or to total intravenous anesthesia. The primary outcome was death from any cause at 1 year. A total of 5400 patients were randomly assigned: 2709 to the volatile anesthetics group and 2691 to the total intravenous anesthesia group. On-pump CABG was performed in 64% of patients, with a mean duration of cardiopulmonary bypass of 79 minutes. The two groups were similar with respect to demographic and clinical characteristics at baseline, the duration of cardiopulmonary bypass, and the number of grafts. At the time of the second interim analysis, the data and safety monitoring board advised that the trial should be stopped for futility. No significant difference between the groups with respect to deaths from any cause was seen at 1 year (2.8% in the volatile anesthetics group and 3.0% in the total intravenous anesthesia group; relative risk, 0.94; 95% confidence interval [CI], 0.69 to 1.29; $P = 0.71$), with data available for 5353 patients (99.1%), or at 30 days (1.4% and 1.3%, respectively; relative risk, 1.11; 95% CI, 0.70 to 1.76), with data available for 5398 patients (99.9%). There were no significant differences between the groups in any of the secondary outcomes or in the incidence of prespecified adverse events, including myocardial infarction. Among patients undergoing elective CABG, anesthesia with a volatile agent did not result in significantly fewer deaths at 1 year than total intravenous anesthesia. (Funded by the Italian Ministry of Health; MYRIAD ClinicalTrials.gov number, NCT02105610.).

[The pharmacokinetic study of desflurane, sevoflurane, isoflurane and enflurane in general anesthesia].

PMID: 11038831 | Indexed in PubMed

To compare pharmacokinetics of desflurane, sevoflurane, isoflurane and enflurane in general anesthesia. 40 patients scheduled for abdominal hysterectomy under general anesthesia were randomly divided into desflurane(D), sevoflurane(S), isoflurane(I) and enflurane(E) groups. After induction of anesthesia and endotracheal intubation, desired fraction (F_d) of desflurane(6%), sevoflurane(2%), isoflurane (1.15%) and enflurane(1.7%) in oxygen and nitrous oxide(1:2) were inhaled in D, S, I and E groups, respectively. The fractional end tidal alveolar concentration (F_a) was adjusted to 1MAC during the maintenance of anesthesia. F_a and the fractional inspired concentration of inhaled anesthetics (F_i) were monitored continuously. During operation, fentanyl was infused continuously and pancuronium was injected intermittently. After the beginning of inhalational anesthesia, the time required for $F_a/F_i = 1:2$ and $F_a = 1MAC$ in D and S groups was significantly shorter than that in E and I groups. The rates of F_a/F_i in D and S groups were significantly higher than those in E and I groups during the maintenance of anesthesia, so were those of F_a/F_d . After cessation of inhalational anesthesia, the time required for F_a equaled to 50% of F_{a0} (the last F_a during stopping administration of the inhalational anesthesia) in D group was

significantly faster than that in the other three groups. The rates of desflurane wash-in and wash-out are faster than those of other inhaled anesthetics. The depth of anesthesia is easy to control when desflurane is used in general anesthesia.
